

SkillUp Hackathon in collaboration with IBM SkillsBuild

AI-based Problem Statement/ Use Cases with IBM Bob

To maximize engagement, learning, and real-world impact, the hackathon is structured into three tracks. Learners can choose any one of the tracks based on their interests, identify a relevant use case or problem statement within that track, and develop an AI powered solution or prototype as their Hackathon project.

1. Student AI Track: Focus on learning outcomes, employability, and student experience.

This track encourages participants to build AI-powered tools that help students learn more effectively, improve productivity, enhance campus experiences, and prepare for future careers. Solutions can address challenges related to education, skill development, career readiness, financial literacy, and student engagement.

2. AI for Impact Track: Focus on societal impact, accessibility, and sustainability.

This track focuses on leveraging AI to address societal and community challenges. Participants are encouraged to develop solutions that promote accessibility, sustainability, digital inclusion, education, healthcare awareness, environment, cybersecurity, and other initiatives that create positive social impact and contribute to a better future.

3. Developer AI Track: Focus on engineering efficiency, automation, and software innovation.

This track is designed for participants interested in using AI to improve the software development lifecycle. Solutions may focus on enhancing developer productivity, automating coding and testing tasks, modernizing legacy systems, streamlining integrations, improving knowledge management, and accelerating onboarding and collaboration within engineering teams.

Student AI Track

1. AI Career Copilot

Why it matters: Many students struggle to identify the right career path and understand the skills required for their dream jobs. AI can help students assess their strengths, identify skill gaps, and receive personalized guidance to improve employability and career readiness.

Use Case: Build an AI career assistant using IBM Bob that:

- Analyzes student skills (resume / GitHub)
- Suggests career paths (AI, cybersecurity, developer, etc.)
- Recommends SkillsBuild courses
- Generates mock interviews + feedback

Project Ideas:

- Resume → AI generates skill gap roadmap
- AI mock interviewer with real-time improvement suggestions
- Personalized “job readiness score”

✓ **Aligns with:** EdTech + HRTech + GenAI copilots

2. AI Study Buddy / Personalized Learning Agent

Why it matters: Every student learns differently. AI-powered learning assistants can provide personalized explanations, generate study plans, and help students learn at their own pace, making education more accessible and effective.

Use Case: Build an AI-powered learning companion using IBM Bob

- Students upload syllabus / notes
 - IBM Bob gives simplified explanations
- IBM Bob creates:
 - Quiz questions (like Kahoot integration)
 - Revision plans

Project Ideas:

- “Explain like I’m 10” learning assistant
- Auto-generate practice tests
- Doubt-solving chatbot trained on course docs

✓ **Aligns with:** Personalized AI + LLM tutoring trend

3. AI-Powered Student Productivity & Time Management

Why it matters: Students often find it difficult to balance coursework, exams, projects, and extracurricular activities. AI can help them organize schedules, prioritize tasks, and build productive study habits.

Use Case: Use IBM Bob to create a productivity assistant

- IBM Bob generates AI planner that:
 - Manages deadlines
 - Tracks study habits
 - Suggests time optimization

Project Ideas:

- “AI semester planner”
- Smart procrastination detector
- Study vs distraction analyzer

✓ **Aligns with:** Personal productivity AI tools trend

4. AI for Campus Life / Hyperlocal Innovation

Why it matters: Campus life involves numerous daily challenges, from finding resources to discovering events and services. AI solutions can improve student experiences by making campus information and services more accessible and efficient.

Use Case: Use IBM Bob to create assistant.

- Campus navigation AI
- Lost & found AI bot
- Smart cafeteria recommendations

Project Ideas:

- AI roommate matcher
- AI event discovery bot
- Smart campus assistant

✓ **Aligns with:** Consumer AI applications

5. AI Gamification Layer (Perfect for scavenger hunt idea)

Why it matters: Engaging students consistently can be challenging. AI-driven gamification can make learning more interactive, motivating, and personalized by adapting challenges and rewards to individual performance

Use Case:

- IBM Bob generates quizzes, challenges, leaderboards
- Adaptive difficulty based on user performance

Project Ideas:

- AI quiz generator (Kahoot-style)
- Hackathon progress gamifier
- AI-generated coding challenges

✅ **Aligns with:** Engagement + gamified learning

6. AI for Financial Literacy (Student-friendly + real-world)

Why it matters: Many students begin managing their finances independently during college but lack financial planning skills. AI can help them understand budgeting, expenses, loans, scholarships, and informed financial decision-making.

Use Case: Use IBM Bob to assist and advise

- Student budget money
- Explain loans
- Track expenses and spending patterns

Project Ideas:

- Student expense analyzer
- “Can I afford this?” AI
- Scholarship & loan advisor

✅ **Aligns with:** FinTech + GenAI advisory tools

AI for Impact Track

1. AI for Healthcare Awareness & Access

Why it matters: Access to reliable healthcare information remains a challenge for many communities, especially in rural and underserved areas. AI can help bridge this gap by providing timely, accurate, and accessible health information, empowering individuals to make informed decisions about their well-being.

Use Case: Build an IBM Bob-powered healthcare assistant that:

- Provides healthcare information in simple language
- Supports multilingual interactions
- Raises awareness about preventive healthcare practices
- Guides users to relevant healthcare resources and services

Project Ideas:

- AI chatbot for rural health awareness
- Maternal and child healthcare information assistant
- Preventive care and vaccination awareness bot
- Multilingual health information companion

✅ **Aligns with:** Healthcare Accessibility + Digital Inclusion + Responsible AI

2. AI for Governance & Citizen Services

Why it matters: Many citizens are unaware of the government schemes and public services available to them or find the information difficult to understand. AI can simplify access to public services and help citizens make better use of available opportunities and benefits.

Use Case: Build an IBM Bob-powered citizen services assistant that:

- Explains government schemes in simple language

- Helps users identify relevant schemes and benefits
- Provides eligibility guidance and application support
- Supports multiple regional languages

Project Ideas:

- Government scheme discovery assistant
- Citizen eligibility checker for welfare programs
- AI-powered public services information bot
- Multilingual governance and policy explainer

✓ **Aligns with:** Public Service Accessibility + Digital Empowerment + Social Impact AI

3. AI for Education & Teacher Empowerment

Why it matters: Teachers spend significant time creating lesson plans, assessments, learning materials, and providing individualized support to students. AI can help reduce administrative workload, personalize instruction, and enable teachers to focus more on student engagement and learning outcomes.

Use Case: Build an IBM Bob-powered teaching assistant that:

- Assists in lesson planning and content creation
- Generates quizzes, assignments, and assessments
- Provides personalized learning recommendations for students
- Supports multilingual content development and classroom engagement

Project Ideas:

- AI lesson plan generator aligned to curriculum objectives
- Automated quiz and assessment creator
- Student performance insights and intervention assistant
- Multilingual content creation and classroom support tool
- AI-powered teaching resource and activity generator

✓ **Aligns with:** Education Innovation + Teacher Productivity + Personalized Learning

4. AI for Climate Action & Sustainability

Why it matters: Climate change and environmental sustainability are among the most pressing global challenges today. AI can help individuals and communities make informed decisions, raise awareness, and promote sustainable practices that contribute to a healthier planet.

Use Case: Build an IBM Bob-powered sustainability assistant that:

- Educates users about climate change and environmental issues
- Promotes sustainable habits and responsible resource usage
- Provides personalized recommendations for reducing environmental impact
- Encourages community participation in sustainability initiatives

Project Ideas:

- Climate awareness and education chatbot
- Personal carbon footprint tracker and advisor
- Waste segregation and recycling assistant
- Sustainable lifestyle recommendation engine
- Community-driven environmental action platform

✓ **Aligns with:** Climate Action + Sustainability + Environmental Awareness + Social Impact

5. Multi-Agent AI Systems (Cutting-edge trend)

Why it matters: Modern AI systems increasingly rely on multiple specialized agents working together to solve complex problems. Building multi-agent applications helps students explore one of the fastest-growing areas of AI innovation.

Use Case: Build multiple IBM Bob agents that collaborate:

- Research agent
- Coding agent

- Testing agent

Project Ideas:

- AI software team simulation
- Autonomous hackathon assistant
- Multi-agent debugging system

✓ **Aligns with:** Agentic AI (AutoGPT-style evolution)

6. AI for Cybersecurity Awareness

Why it matters: As digital adoption increases, cyber threats are becoming more common. AI can help individuals identify risks, recognize fraudulent activities, and build safer online habits through accessible security education.

Use Case: IBM Bob helps users by

- identifying phishing messages
- explaining security risks simply

Project Ideas:

- “Is this safe?” AI analyzer
- AI-powered password strength + behavior advisor
- Cyber awareness chatbot for beginners

✓ **Aligns with:** Security + AI convergence

7. Multilingual AI Assistant (India-focused impact)

Why it matters: Language barriers often limit access to education, information, and digital services. AI-powered multilingual assistants can make knowledge and technology accessible to a broader and more diverse population.

Use Case:

- IBM Bob translates + explains technical content in regional languages

Project Ideas:

- English → Hindi learning bot
- Technical concept simplifier in local languages
- AI subtitle generator for courses

✓ **Aligns with:** Inclusive AI + localization

Developer AI Track

1. AI-Driven Code Intelligence and Automated Documentation System

Why it matters: Developers spend significant time understanding unfamiliar code and maintaining documentation. AI can accelerate software development by automatically explaining code, generating documentation, and improving knowledge transfer

Use Case examples:

- Upload code → IBM Bob explains what the code does
- IBM Bob creates architecture diagrams automatically
- IBM Bob generates API documentation

Project Ideas:

- Tool that converts code into simple explanations
- AI assistant that explains each file in a project
- Visual diagram generator for system design

✓ **Aligns with:** AI copilots, productivity tools, automated documentation

2. Intelligent AI-Powered Test Generation and Quality Assurance Framework

Why it matters: Software quality depends heavily on effective testing, but creating comprehensive test cases is often time-consuming. AI can automate test generation, identify edge cases, and improve application reliability.

Use Case examples:

- Input code → IBM Bob creates test cases
- IBM Bob finds edge cases automatically

- IBM Bob generates test scenarios

Project Ideas:

- AI tool that creates unit test cases
- Test generator for APIs
- Smart bug detection assistant

✓ **Aligns with:** AI in testing, quality improvement, automation

3. AI-Assisted Legacy Application Modernization and Transformation Platform

Why it matters: Many organizations rely on legacy systems that are expensive and difficult to maintain. AI can simplify modernization efforts by recommending improvements, refactoring code, and supporting migration to newer technologies.

Use Case examples:

- Upload old code → IBM Bob suggests improvements
- IBM Bob recommends better system design
- IBM Bob helps convert code into newer frameworks

Project Ideas:

- Tool that transforms old code into modern code
- AI assistant for code refactoring
- Migration planner for legacy systems

✓ **Aligns with:** modernization, code transformation, enterprise systems

4. Smart API Integration and AI-Ready Connectivity Framework

Why it matters: Modern applications depend on multiple systems working together. AI can simplify integration processes, reduce development effort, and help organizations connect services more efficiently.

Use Case examples:

- Input API details → IBM Bob creates integration code
- IBM Bob builds connectors between systems

- IBM Bob generates API usage documentation

Project Ideas:

- API integration generator
- Tool that connects multiple services automatically
- AI-based API documentation builder

✓ **Aligns with:** API development, system integration, automation

5. AI-Powered Knowledge Management and Context-Aware Learning Hub

Why it matters: Finding the right information quickly is a common challenge for teams and organizations. AI-powered knowledge systems can improve productivity by delivering relevant information and guidance when needed.

Use Case examples:

- User asks a question → IBM Bob gives relevant answer
- IBM Bob searches company documents
- IBM Bob provides step-by-step guidance

Project Ideas:

- AI chatbot for internal documentation
- Knowledge search assistant
- Developer Q&A bot

✓ **Aligns with:** AI assistants, knowledge sharing, search systems

6. Intelligent AI Onboarding and Developer Assistance Platform

Why it matters: New developers often require significant time to understand project structures, tools, and workflows. AI assistants can accelerate onboarding by providing contextual guidance, learning resources, and instant support.

Use Case examples:

- New user interacts with IBM Bob → learns system step-by-step
- IBM Bob explains project structure

- IBM Bob answers development questions

Project Ideas:

- AI onboarding assistant
- Interactive learning chatbot
- Developer help assistant

✓ **Aligns with:** learning tools, AI copilots, productivity improvement
